

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

```
df = pd.read_excel('/content/marketing_campaign.xlsx')
```

```
df.head()
```

	ID	Year_Birth	Education	Marital_Status	Income	Kidhome	Teenhome	Dt
0	5524	1957	Graduation	Single	58138.0	0	0	
1	2174	1954	Graduation	Single	46344.0	1	1	
2	4141	1965	Graduation	Together	71613.0	0	0	
3	6182	1984	Graduation	Together	26646.0	1	0	
4	5324	1981	PhD	Married	58293.0	1	0	

5 rows × 29 columns

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 2240 entries, 0 to 2239
Data columns (total 29 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   ID                                    2240 non-null   int64
1   Year_Birth                           2240 non-null   int64
2   Education                             2240 non-null   object
3   Marital_Status                       2240 non-null   object
4   Income                                2216 non-null   float64
5   Kidhome                              2240 non-null   int64
6   Teenhome                             2240 non-null   int64
7   Dt_Customer                          2240 non-null   object
8   Recency                              2240 non-null   int64
9   MntWines                             2240 non-null   int64
10  MntFruits                             2240 non-null   int64
11  MntMeatProducts                       2240 non-null   int64
12  MntFishProducts                       2240 non-null   int64
13  MntSweetProducts                      2240 non-null   int64
14  MntGoldProds                          2240 non-null   int64
15  NumDealsPurchases                     2240 non-null   int64
16  NumWebPurchases                       2240 non-null   int64
17  NumCatalogPurchases                   2240 non-null   int64
18  NumStorePurchases                     2240 non-null   int64
19  NumWebVisitsMonth                     2240 non-null   int64
20  AcceptedCmp3                          2240 non-null   int64
21  AcceptedCmp4                          2240 non-null   int64
22  AcceptedCmp5                          2240 non-null   int64
23  AcceptedCmp1                          2240 non-null   int64
```

```
24 AcceptedCmp2      2240 non-null   int64
25 Complain         2240 non-null   int64
26 Z_CostContact     2240 non-null   int64
27 Z_Revenue         2240 non-null   int64
28 Response          2240 non-null   int64
dtypes: float64(1), int64(25), object(3)
memory usage: 507.6+ KB
```

```
df.isnull().sum()
```

	0
ID	0
Year_Birth	0
Education	0
Marital_Status	0
Income	24
Kidhome	0
Teenhome	0
Dt_Customer	0
Recency	0
MntWines	0
MntFruits	0
MntMeatProducts	0
MntFishProducts	0
MntSweetProducts	0
MntGoldProds	0
NumDealsPurchases	0
NumWebPurchases	0
NumCatalogPurchases	0
NumStorePurchases	0
NumWebVisitsMonth	0
AcceptedCmp3	0
AcceptedCmp4	0
AcceptedCmp5	0
AcceptedCmp1	0
AcceptedCmp2	0
Complain	0
Z_CostContact	0
Z_Revenue	0
Response	0

dtype: int64

```
df = df.dropna()
```

```
df = df.drop_duplicates()
```

```
df.columns
```

```
Index(['ID', 'Year_Birth', 'Education', 'Marital_Status', 'Income',
      'Kidhome',
      'Teenhome', 'Dt_Customer', 'Recency', 'MntWines', 'MntFruits',
      'MntMeatProducts', 'MntFishProducts', 'MntSweetProducts',
      'MntGoldProds', 'NumDealsPurchases', 'NumWebPurchases',
      'NumCatalogPurchases', 'NumStorePurchases', 'NumWebVisitsMonth',
      'AcceptedCmp3', 'AcceptedCmp4', 'AcceptedCmp5', 'AcceptedCmp1',
      'AcceptedCmp2', 'Complain', 'Z_CostContact', 'Z_Revenue',
      'Response'],
      dtype='object')
```

```
df['Dt_Customer'] = pd.to_datetime(df['Dt_Customer'])
```

```
df['Total_Spending'] = (
    df['MntWines'] +
    df['MntFruits'] +
    df['MntMeatProducts'] +
    df['MntFishProducts'] +
    df['MntSweetProducts'] +
    df['MntGoldProds']
)
```

```
df.describe()
```

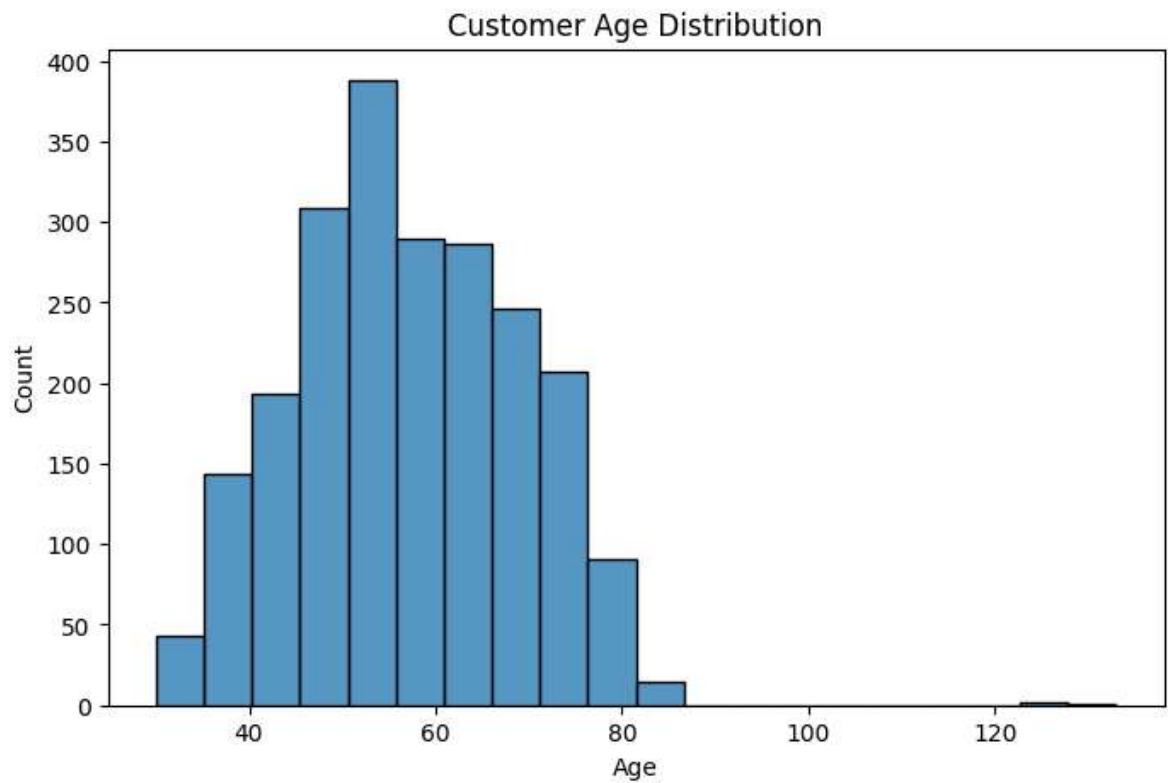
	ID	Year_Birth	Income	Kidhome	Teenhome
count	2216.000000	2216.000000	2216.000000	2216.000000	2216.000000
mean	5588.353339	1968.820397	52247.251354	0.441787	0.505415
min	0.000000	1893.000000	1730.000000	0.000000	0.000000
25%	2814.750000	1959.000000	35303.000000	0.000000	0.000000
50%	5458.500000	1970.000000	51381.500000	0.000000	0.000000
75%	8421.750000	1977.000000	68522.000000	1.000000	1.000000
max	11191.000000	1996.000000	666666.000000	2.000000	2.000000
std	3249.376275	11.985554	25173.076661	0.536896	0.544181

8 rows × 28 columns

```
plt.figure(figsize=(8,5))
sns.histplot(2026 - df['Year_Birth'], bins=20)

plt.title('Customer Age Distribution')
plt.xlabel('Age')
plt.ylabel('Count')

plt.show()
```

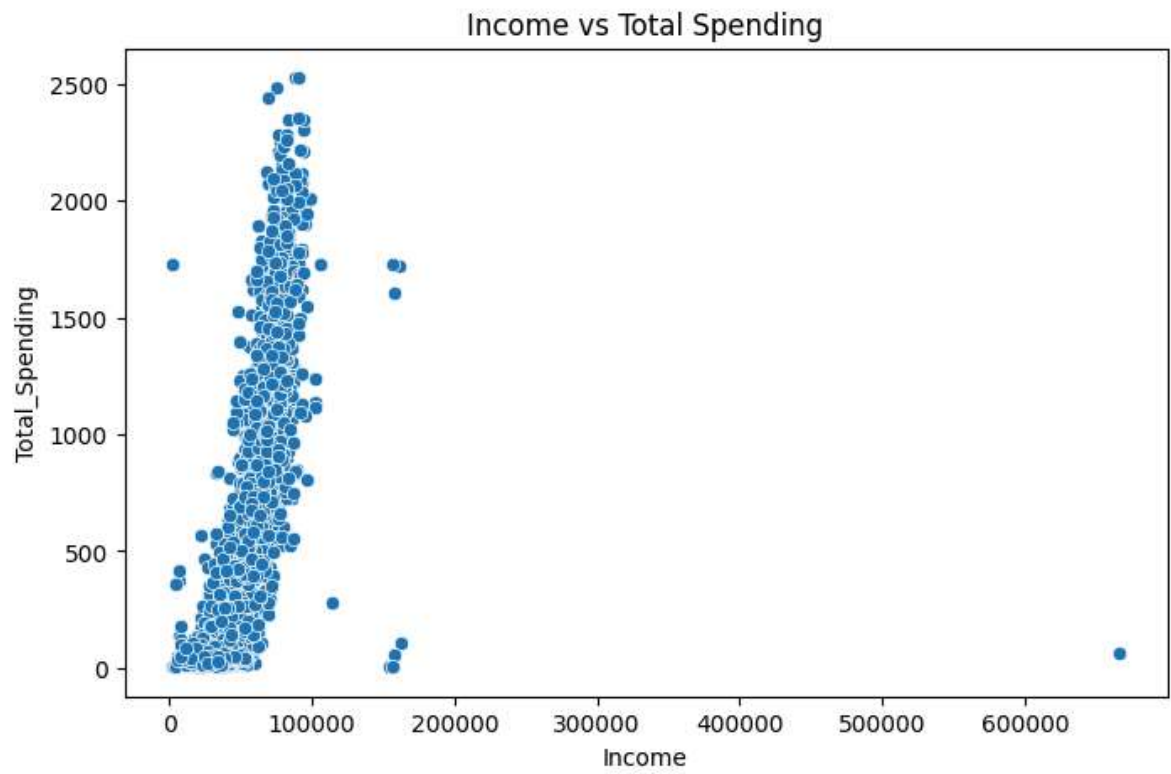


```
plt.figure(figsize=(8,5))

sns.scatterplot(
    x='Income',
    y='Total_Spending',
    data=df
)

plt.title('Income vs Total Spending')

plt.show()
```



```
channels = [  
    'NumWebPurchases',  
    'NumCatalogPurchases',  
    'NumStorePurchases'  
]  
  
df[channels].sum().plot(  
    kind='bar',  
    figsize=(7,5)  
)  
  
plt.title('Purchases by Channel')  
plt.ylabel('Total Purchases')  
  
plt.show()
```



```
response_counts = df['Response'].value_counts()

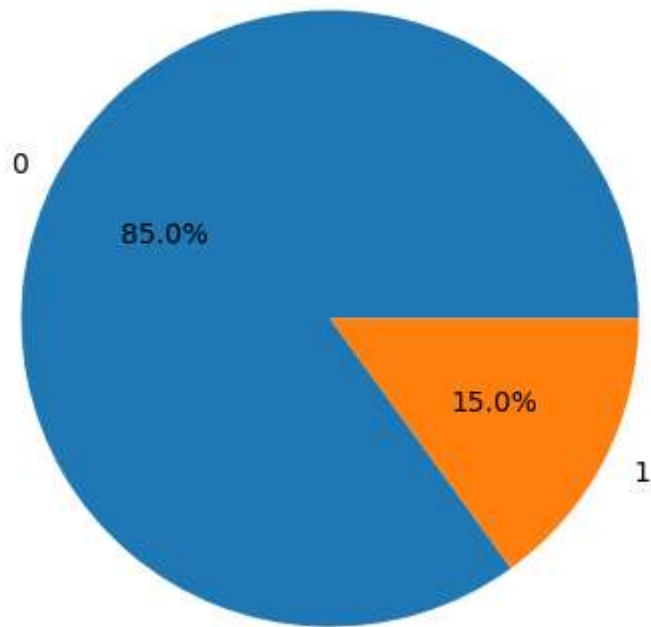
plt.figure(figsize=(5,5))

plt.pie(
    response_counts,
    labels=response_counts.index,
    autopct='%1.1f%%'
)

plt.title('Campaign Response')

plt.show()
```

Campaign Response



```
products = [  
    'MntWines',  
    'MntFruits',  
    'MntMeatProducts',  
    'MntFishProducts',  
    'MntSweetProducts'  
]  
  
df[products].sum().plot(  
    kind='bar',  
    figsize=(8,5)  
)  
  
plt.title('Product Category Spending')  
plt.ylabel('Amount Spent')  
  
plt.show()
```